Linear Discriminant Functions and Fisher Linear Discriminant

Basic framework

- Suppose we are dealing with a classification problem with K classes. Assume that each instance is represented as a vector $\mathbf{x} \in \mathbb{R}^d$.
- Idea: Use a **linear function** of input features as a hyperplane to separate classes.

Advantages and use case scenarios

- Solution has a closed-form expression in many situations and the computation overhead is usually low.

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- Solution has a closed-form expression in many situations and the computation overhead is usually low.
- Works well when classes are approximately Gaussian with similar covariance.

Linear Discriminant Function

A linear discriminant function is defined as:

$$g(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$$

where:

- $\mathbf{w}$ = weight vector
- w_0 = bias term

Decision rule (binary classification):

$$\text{Class} \begin{cases} \mathcal{C}_1 & \text{if } g(\mathbf{x}) > 0 \\ \mathcal{C}_2 & \text{if } g(\mathbf{x}) < 0 \end{cases}$$

Geometric Interpretation

- Decision boundary:

$$\mathbf{w}^T\mathbf{x} + w_0 = 0$$

  represents a **hyperplane** and **w** is perpendicular to the decision boundary.

- Signed distance from boundary:

$$\frac{\mathbf{w}^T\mathbf{x} + w_0}{\|\mathbf{w}\|}$$

An illustration

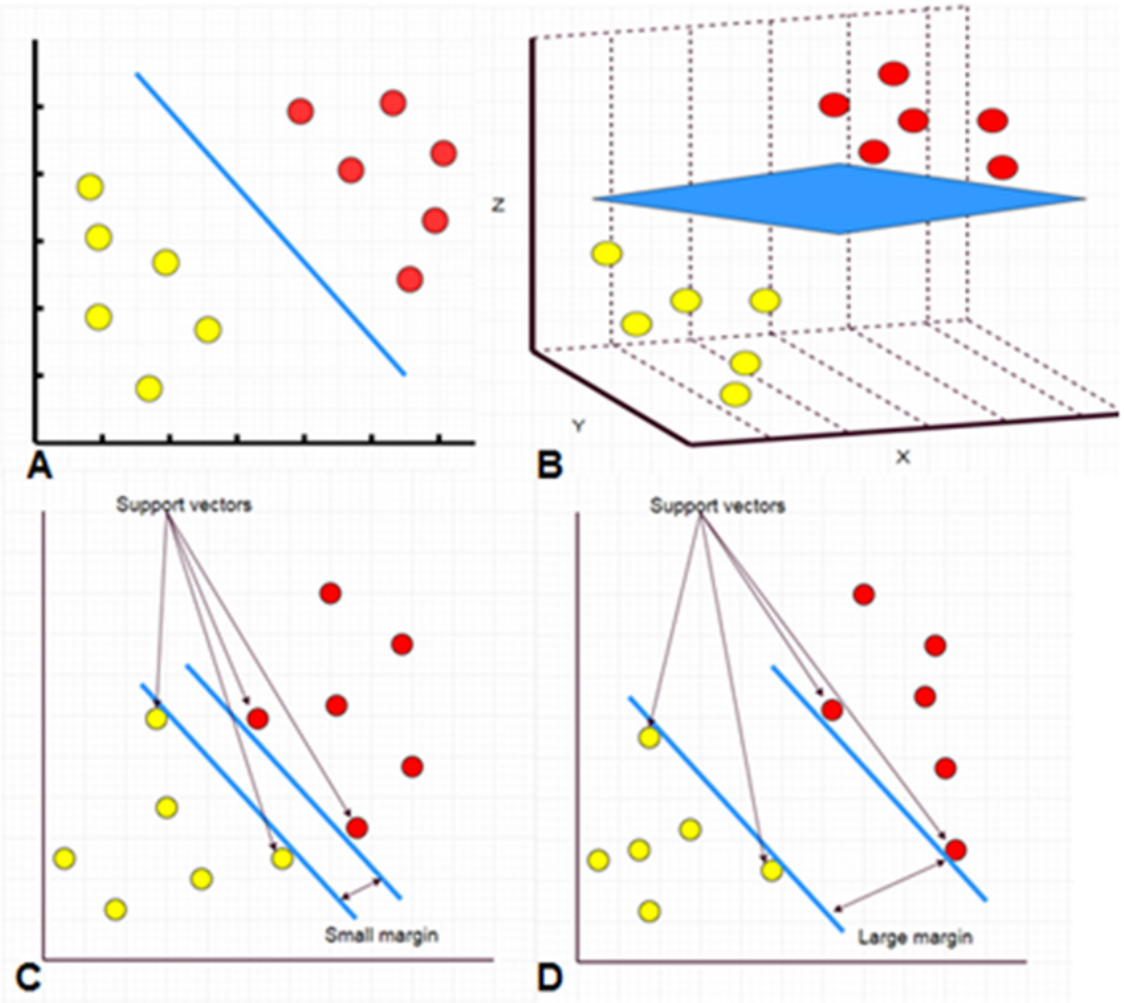


Figure: Illustration of hyperplane separating two classes

Two-Class Problem Setup

Assume:

- Class means: $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$
- Within-class scatter:

$$S_W = S_1 + S_2$$

$$S_i = \sum_{\mathbf{x} \in \mathcal{C}_i} (\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T$$

Fisher's Idea

Project data onto a direction $\mathbf{w}$:

$$y = \mathbf{w}^T \mathbf{x}$$

so as to:

- Maximize separation of projected means
- Minimize projected variance

Fisher Criterion

Projected means:

$$m_i = \mathbf{w}^T \boldsymbol{\mu}_i$$

Projected variance:

$$s_i^2 = \mathbf{w}^T S_i \mathbf{w}$$

Fisher's objective function:

$$J(\mathbf{w}) = \frac{(\mathbf{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))^2}{\mathbf{w}^T S_W \mathbf{w}}$$

Optimal Direction

Maximizing $J(\mathbf{w})$ gives:

$$\mathbf{w} \propto S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

Key insight:

- Direction depends on difference of means
- Scaled by inverse within-class scatter

Algorithm Steps

1. Compute class means $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$
2. Compute within-class scatter matrix S_W
3. Compute
$$\mathbf{w} = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$
4. Project data: $y = \mathbf{w}^T\mathbf{x}$
5. Choose threshold between projected class means

A simple numerical example

Two classes in $\mathbb{R}^2$

Class $\mathcal{C}_1$:

$$(2,3),\ (3,3),\ (2,2)$$

Class $\mathcal{C}_2$:

$$(7,8),\ (8,8),\ (7,7)$$

Goal: Find Fisher's optimal projection direction.

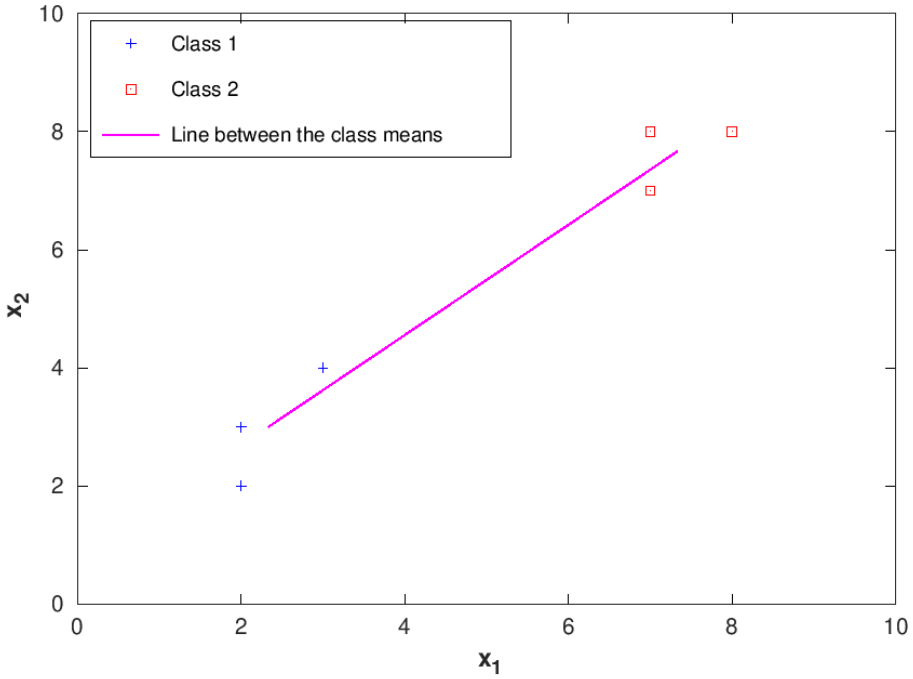


Figure: A simple illustrative 2D example

Step 1: Compute Class Means

$$\boldsymbol{\mu}_1 = \frac{1}{3}\begin{bmatrix} 2+3+2 \\ 3+3+2 \end{bmatrix} = \begin{bmatrix} 2.33 \\ 2.67 \end{bmatrix}$$

$$\boldsymbol{\mu}_2 = \frac{1}{3}\begin{bmatrix} 7+8+7 \\ 8+8+7 \end{bmatrix} = \begin{bmatrix} 7.33 \\ 7.67 \end{bmatrix}$$

Mean difference:

$$\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \begin{bmatrix} -5 \\ -5 \end{bmatrix}$$

Step 2: Within-Class Scatter

For class $\mathcal{C}_1$:

$$S_1 = \sum (\mathbf{x} - \mu_1)(\mathbf{x} - \mu_1)^T = \begin{bmatrix} 0.67 & 0.33 \\ 0.33 & 0.67 \end{bmatrix}$$

For class $\mathcal{C}_2$:

$$S_2 = \begin{bmatrix} 0.67 & 0.33 \\ 0.33 & 0.67 \end{bmatrix}$$

Total:

$$S_W = S_1 + S_2 = \begin{bmatrix} 1.34 & 0.66 \\ 0.66 & 1.34 \end{bmatrix}$$

Step 3: Compute Optimal Direction

Fisher solution:

$$\mathbf{w} = S_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

$$S_W^{-1} = \begin{bmatrix} 1 & -0.49 \\ -0.49 & 1 \end{bmatrix} \quad \text{(approx)}$$

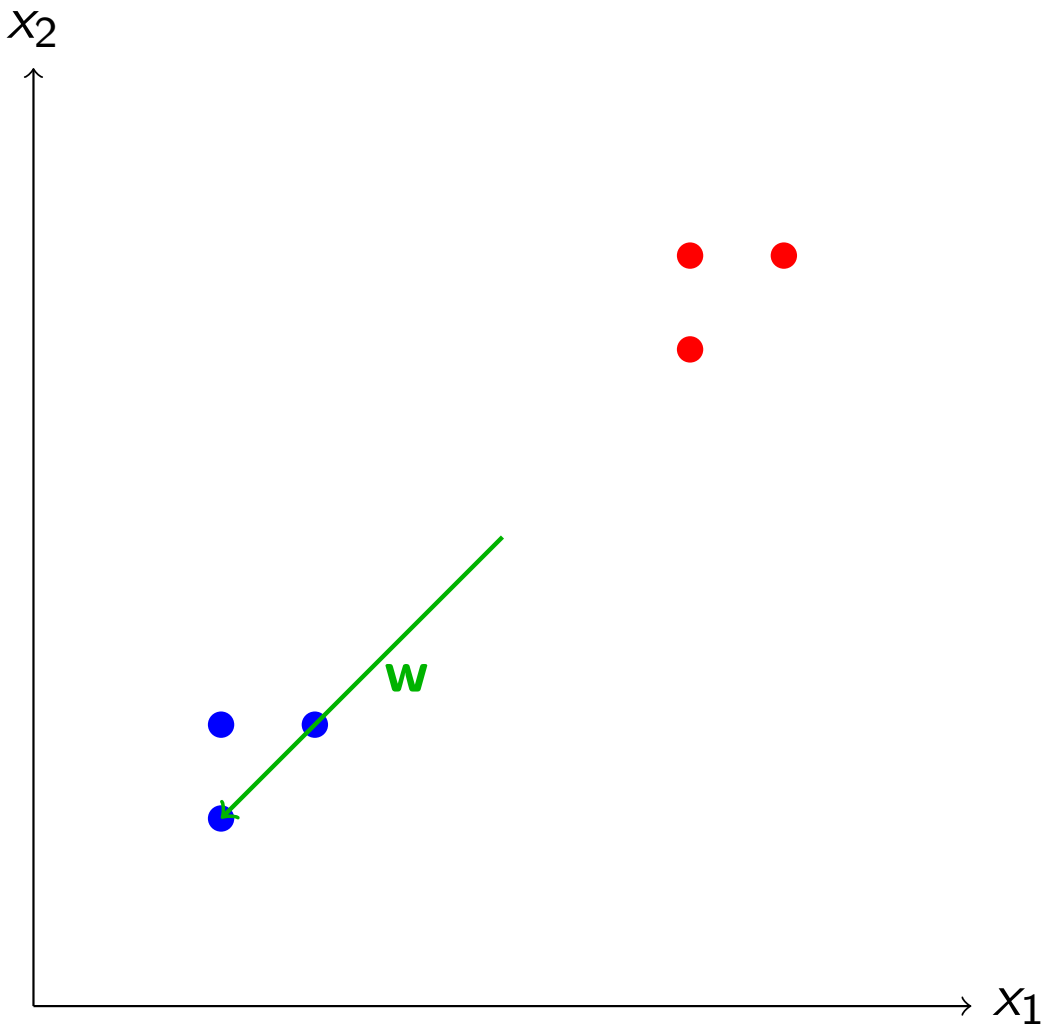
$$\mathbf{w} \propto \begin{bmatrix} -2.55 \\ -2.55 \end{bmatrix}$$

Normalized direction:

$$\mathbf{w} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

Direction lies along line $x_1 + x_2$.

Illustration of Data and Direction



Blue $= \mathcal{C}_1$, Red $= \mathcal{C}_2$

Projection onto $\mathbf{w}$

Projection formula:

$$y = \mathbf{w}^T \mathbf{x}$$

For $\mathcal{C}_1$:

$$y \approx -\frac{1}{\sqrt{2}}(x_1 + x_2)$$

Class 1 projected values are clustered around:

$$-3.5$$

Class 2 projected values are clustered around:

$$-10.6$$

Clear separation on 1D axis.

Key Observations

- Projection direction aligns with mean difference.
- Within-class scatter slightly adjusts direction.
- After projection, classification becomes 1D thresholding.
- Fisher LDA reduces dimensionality while preserving separability.

Extension to Multiple Classes

Between-class scatter:

$$S_B = \sum_{k=1}^{K} N_k(\boldsymbol{\mu}_k - \boldsymbol{\mu})(\boldsymbol{\mu}_k - \boldsymbol{\mu})^T$$

Optimization:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$$

Leads to generalized eigenvalue problem:

$$S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$$